Phase 2 — Technology Stack

Source-grounded standalone document

Technology Stack

Category	Technology	Documented Use
Runtime	Node.js v18+	Server runtime and routing infrastructure.
Package Manager	npm v9+	Operational dependency management.
Backend	Express.js v4/v5	RESTful API routing.
Database	MongoDB	Primary NoSQL data store.
ODM	Mongoose	Data modeling, validation and MongoDB queries.
Security	jsonwebtoken	JWT authentication and route guarding.
Security	bcrypt	Password hashing.
AI Integration	Google Gemini SDK / @google/genai	AI question-answer and topic-generation workflows.
API Testing	Postman / ThunderClient	Endpoint and schema validation testing.
Development	Visual Studio Code or similar	Development environment.

The source also specifies Windows 10/11, macOS or Linux, minimum 8 GB RAM, and 1 GB available disk workspace.